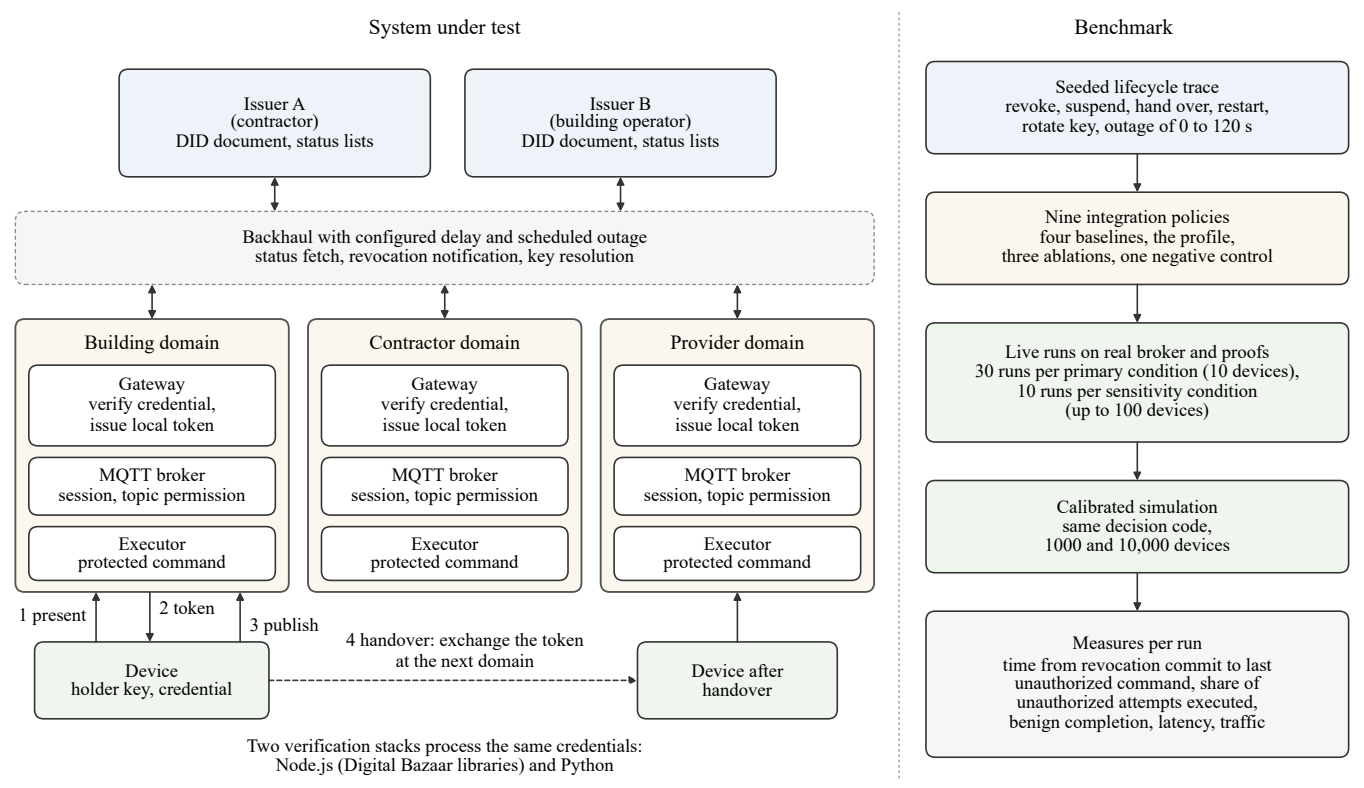
System under test
Issuer A
(contractor)
DID document, status lists
Issuer B
(building operator)
DID document, status lists
Backhaul with configured delay and scheduled outage
status fetch, revocation notification, key resolution
Building domain
Gateway
verify credential,
issue local token
MQTT broker
session, topic permission
Executor
protected command
Contractor domain
Gateway
verify credential,
issue local token
MQTT broker
session, topic permission
Executor
protected command
Provider domain
Gateway
verify credential,
issue local token
MQTT broker
session, topic permission
Executor
protected command
1 present
2 token
3 publish
Device
holder key, credential
4 handover: exchange the token
at the next domain
Device after
handover
Two verification stacks process the same credentials:
Node.js (Digital Bazaar libraries) and Python
Benchmark
Seeded lifecycle trace
revoke, suspend, hand over, restart,
rotate key, outage of 0 to 120 s
Nine integration policies
four baselines, the profile,
three ablations, one negative control
Live runs on real broker and proofs
30 runs per primary condition (10 devices),
10 runs per sensitivity condition
(up to 100 devices)
Calibrated simulation
same decision code,
1000 and 10,000 devices
Measures per run
time from revocation commit to last
unauthorized command, share of
unauthorized attempts executed,
benign completion, latency, traffic